higher education & training

Department:
Higher Education and Training
REPUBLIC OF SOUTH AFRICA

NATIONAL CERTIFICATE (VOCATIONAL)

SYSTEM ANALYSIS AND DESIGN
NQF LEVEL 3

(10041003)

4 March 2019 (X-Paper)
09:00–12:00

This question paper consists of 10 pages.

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TIME: 3 HOURS
MARKS: 200

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions.
2. Read ALL the questions carefully.
3. Number the answers according to the numbering system used in this question paper.
4. Write neatly and legibly.

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SECTION A

QUESTION 1

1.1 Various options are given as possible answers to the following questions. Choose the answer and write only the letter (A–D) next to the question number (1.1.1–1.1.10) in the ANSWER BOOK.

1.1.1 Which ONE of the following statements best defines a *project*?

A Planned activity consisting of defined tasks by a team of people
B Sequence of steps with a scheduled completion
C Set of procedures that must be completed in a timely fashion
D Planned undertaking with a start and stop that produces a result

1.1.2 The organisation and direction of people to achieve a planned result within a predetermined schedule and budget:

A Project delivery management
B Project integration management
C Project human resource management
D Project management

1.1.3 Also known as the breakeven point:

A Net present value (NPV)
B Payback period
C Return on investment (ROI)
D Cost/benefit equalisation

1.1.4 The … is an important factor of management information system.

A system
B data
C process
D information

1.1.5 The ... level supplies information to the strategic tier for the use of top management.

A operational
B environmental
C competitive
D tactical

1.1.6 One of the following can be defined as data processed into a meaningful form for the recipient and of real or perceived value in current or prospective decisions:

A System
B Information
C Technology
D Service

1.1.7 In the analysis phase the development of the … occurs, which is a clear statement of the goals and objectives of the project.

A documentation
B flow chart
C program specifications
D design

1.1.8 Technical writers generally provide the … for the new system.

A programs
B network
C analysis
D documentation

1.1.9 … design and implement database structures.

A Programmers
B Project managers
C Technical writers
D Database administrators

1.1.10 … manage the system development, assign staff, manage the budget and reporting, and ensure that deadlines are met.

A Project managers
B Network engineers
C Graphic designers
D Systems analysts

(10 × 1) (10)

1.2 Indicate whether the following statements are TRUE or FALSE. Choose the correct answer and write only 'True' or 'False' next to the question number (1.2.1–1.2.10) in the ANSWER BOOK.

1.2.1 Information systems software focuses on development, consulting and training of staff.

1.2.2 A competitive advantage means an organisation makes less money than its rivals.

1.2.3 The system development life cycle is an organisational process of developing and maintaining a system.

1.2.4 In the system analysis and design terminology the system development life cycle refers to the software development life cycle.

1.2.5 An accounting information system is an information system that automates the role of accounting in an organisation.

1.2.6 System analysis and system design are the same phase of the system development life cycle.

1.2.7 The budget of a company is usually done on a monthly basis.

1.2.8 A forensic audit is a formal investigation to identify theft or fraud.

1.2.9 Coding is not a step in the system development life cycle.

1.2.10 A collection of components that work together to realise some objectives forms a system.

(10 × 1) (10)

1.3 Write the following abbreviations in full:

1.3.1 RFQ

1.3.2 BPA

1.3.3 OAS

1.3.4 DBMS

1.3.5 BPI

1.3.6 BPR

1.3.7 ROI

1.3.8 SWOT

1.3.9 EIS

1.3.10 WAN

(10 × 1) (10)

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1.4 Complete the following sentences by filling in the missing word or words. Write only the word or words next to the question number (1.4.1–1.4.10) in the ANSWER BOOK.

1.4.1 A system that processes financial data is called an …

1.4.2 … gives an organisation full use of the computer technology for an agreed time at an agreed monthly cost.

1.4.3 The … is the head of the information technology or information systems department of an organisation.

1.4.4 … are guidelines for the adoption, management, use and support of technologies needed for the organisational plan.

1.4.5 … is a way in which a comparison is made between two or more performances of the same process.

1.4.6 … is the way in which organisations plan for the future.

1.4.7 … is the study of the health and comfort factors associated with users who spend a lot of time at their desks working with computers.

1.4.8 The … is responsible for information systems architecture and manages emerging technologies in terms of research, planning and adoption.

1.4.9 … are designed to provide interactive ad hoc support for tactical decision-making.

1.4.10 A/An … is a collection of people, machines, materials and procedures that work together to produce a specified product or results.

(10 × 1) (10)
[40]

TOTAL SECTION A: 40

SECTION B

QUESTION 2

2.1 Name SIX different information systems development methodologies. (6)

2.2 List SEVEN stages of the systems development life cycle (SDLC). (7)

2.3 Give TWO motivational factors that may lead to the development of an information system. (2)

[15]

QUESTION 3

An accounting information system (AIS) is a structure that a business uses to collect, store, manage, process, retrieve and report its financial data so that it can be used by accountants, consultants, business analysts, managers, chief financial officers auditors and regulatory and tax agencies. (www.investopedia.com)

3.1 Define the term *data processing*. (2)

3.2 Identify FOUR control systems within an organisation. (4)

3.3 Differentiate between a *knowledge management system, a transaction processing system* and a *decision support system.* (6)

3.4 Differentiate between *enterprise resource planning* and *accounting information system*. (2 × 2) (4)

[16]

QUESTION 4

4.1 Explain the term *strategic information systems management*. (2)

4.2 There are different information system professionals in an organisation.

Give the roles of each of the following professionals:

4.2.1 Chief network officer

4.2.2 Chief information security officer

4.2.3 Chief knowledge officer

4.2.4 Chief technology officer

(4 × 2) (8)

4.3 Identify SEVEN tasks within an information systems department. (7)

4.4 Differentiate between *decentralisation* and *centralisation* within an organisation. (4)

4.5 Draw an organisational diagram to show the information systems department and its sub-departments. (7)
[28]

QUESTION 5

As the IT assistant director of the Limpopo TVET College you have to recommend various methods of financing new computers for the college.

5.1 Give THREE advantages and THREE disadvantages of in-house development. (6)

5.2 Explain the importance of standardisation in terms of hardware or software during the acquisition of new computer technology and give an example. (3)

5.3 Identify FIVE factors that affect the location of a computer which is installed for the first time. (5)

5.4 Differentiate between *information systems off-shoring* and *information systems outsourcing.* (4)

5.5 Identify SIX steps to follow when acquiring new computer technology. (6)

5.6 Discuss THREE methods or options for financing the acquisition of computer technology. (3 × 2) (6)
[30]

QUESTION 6

As the technical advisor of Thohoyandou Computer Corporation you have to advise the company on external environmental factors, including risks and threats of competitors who are offering the same business.

6.1 Define the following terms:

6.1.1 Cash cow

6.1.2 Star

6.1.3 Dog

6.1.4 Problem child

6.1.5 Positional analysis

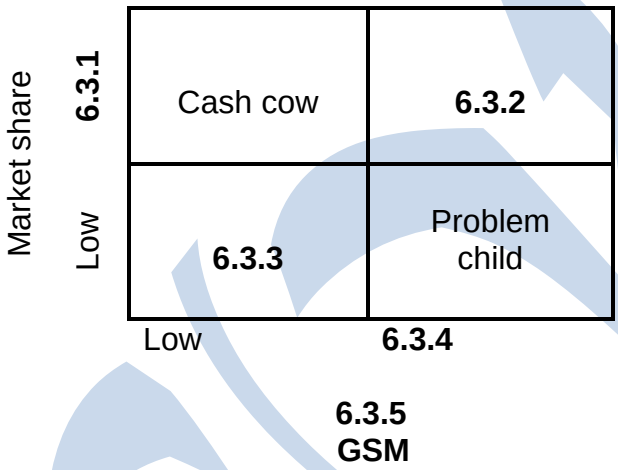
6.1.6 Applications portfolio

(6 × 2) (12)

6.2 Differentiate between *systems portfolio* and *company portfolio* by means of a table. (4)

6.3 The Boston Consulting Group developed the growth share matrix (GSM) to help companies analyse portfolios or products in terms of their market share and the level of growth in the market. This helps organisations to plan strategies involving ongoing investment in products.

Complete the following structure by writing the missing answer next to the question number (6.3.1–6.3.5) in the ANSWER BOOK.



(5 × 1) (5)

6.4 6.4.1 Identify THREE activities of the operational stage of information systems planning. (3)

6.4.2 Identify THREE activities of the tactical stage of information systems planning. (3)

6.4.3 Identify FOUR guiding objectives of strategic information systems planning. (4)

6.5 Draw a diagram that shows Michael Porter's Five Forces Model according to the theoretical frameworks for organisational and information systems planning. (8)

[39]

QUESTION 7

You are a business analyst and have to develop a system for a small company.

7.1 Name and briefly explain FIVE different approaches used to gather data. (10)

7.2 Give THREE differences each between *root cause analysis* and *problem analysis*. (3 × 2) (6)

7.3 State the SIX steps involved in cost-benefit analysis. (6)

7.4 Below is a data table on investment with some information missing.

TASKS	TOTAL	YEAR 01	YEAR 02	YEAR 03	YEAR 04
Software development cost	R5 000 000,00	R5 000 000,00			
Deployment and maintenance costs	**A?**	R1 250 000,00	R960 000,00	R570 000,00	R310 000,00
TOTAL COSTS	R8 090 000,00	**B?**	R960 000,00	R570 000,00	**C?**
Present value of benefits – tangible	R9 050 000,00	R750 000,00	**D?**	R2 700 000,00	R3 900 000,00
Overall profits/losses	R960 000,00	**E?**	R740 000,00	R2130000,00	R3 590 000,00

DATA TABLE

7.4.1 Complete the table by writing the missing amounts next to the letters (A–E) in the ANSWER BOOK. (5)

7.4.2 Calculate the return on investment (ROI) of this project using the data in the table. (5)

[32]

TOTAL SECTION B: 160
GRAND TOTAL: 200

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